EXP no:2 MATRIX CREATION AND BASIC
30/1/26 MATRIX OPERATION IN MATLAB

Aim:-

Generate a matrix and perform Basic operation on Matrices using MATLAB Software.

THEORY:-

MATLAB Treats all variables as Matrices vectors are special forms of Matrices and contain only one row or one coulomb whereas Scalars are special forms of matrices and contain only one row and one coulmn. A matrix with one row vector and a Matrix with single coulmn is called coulmn vector.

PROCEDURE:-

1. open MATLAB software, The MATLAB window opens with Command window, workspace etc..

2. In the Command window Create single row vector, Column vector, press enter to view the output.

3. Next Create two-dimensional matrices and perform matrix operations.

4. perform Addition, subtraction, multiplication, driven, transpose and inverse operation on the Matrices.

EXP NO 13
12/2/h

SIGNAL OPERATIONS : FOLDING, ENERGY AND AVERAGE POWER COMPUTATION

AIM :-

To perform functions on Signals and Sequences such as folding, computation of energy and average power.

SOFTWARE REQUIRED :-

MATLAB Software.

THEORY :-

Signal Addition :-

Any two signals can be added to form a third signal,

$$z(t) = x(t) + y(t)$$

Multiplication :-

Multiplication of two signals can be obtained by Multiplying their values at every instant z,

$$z(t) = x(t)\,y(t).$$

Time Reversal / Folding :-

Time Reversal of signal x(t) can be obtained by folding the signal about t = 0.

$$Y(t) = y(-t)$$

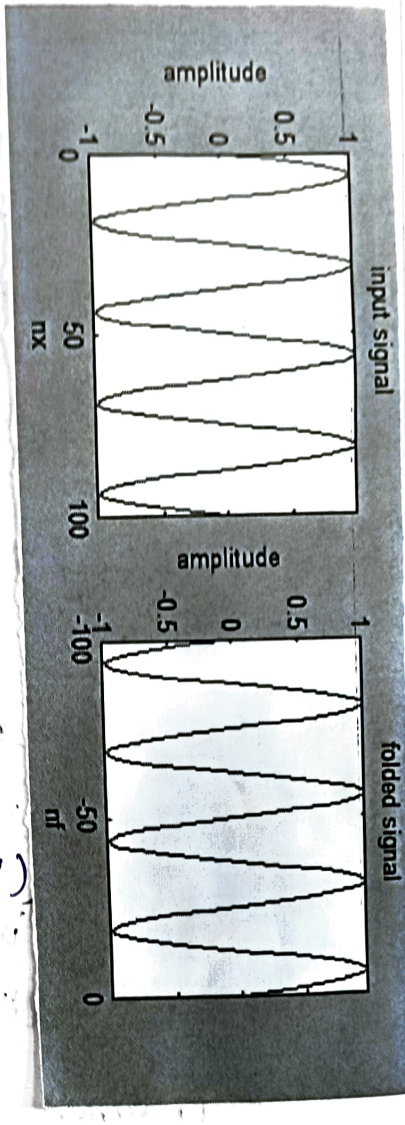
Signal Amplification / Scaling :-

% Scaling of a Signal

```
A= 2;
y3= A*x1;
figure;
subplot (2,2,1);
plot(t,x1);
xlabel('time');
Ylabel('Amplitude');
title('Amplitude input Signal');
```

% folding of a Signal

```
h= length (x1);
nx = 0:h-1;
subplot (2,2,3);
plot (nx, x1);
xlabel('nx');
Ylabel('amplitude');
title ('Amplified input Signal')
y4 = fliplr(x1);
nf = fliplr(nx);
subplot (2,2,4);
plot (nf, y4);
xlabel ('nf');
Ylabel ('Amplitude');
title ('folded Signal');
```



% Shifting of a Signal

```
figure;
subplot (3,1,1);
plot (t,x1);
xlabel ('time t');
Ylabel ('Amplitude');
title ('input signal');
subplot (3,1,2);
```

$y(x) = a\,x(n)$ if $a<1$ attenuation

$a>1$ amplification.

Time Shifting:-

The time Shifting of $x(n)$ obtained by delay or advance the Signal in time by using

$$y(n) = x(n+k)$$

If k is a positive number, $y(x)$ it gets shifted to the Right

i.e the shifting delays the signals.

If k is a Negative number, $y(x)$ it gets shifted left ∴ Signal shifting advances the Signal.

EXP: 4

19/2/16

Plotting Magnitude and phase Response of a Given System.

Aim:-

Write a MATLAB program to plot Magnitude and phase response of a given System.

Software Require :-

MATLAB 7.9

PROCEDURE:-

* open MATLAB
* Open new M-file.
* Type the program.
* Save in current directory
* Compile and Run the program
* For the output See command window Figure window.

Theory:-

If $X(j\Omega)$ is fourier Transform of Continuous time Signal $x(t)$, then

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt \quad \text{for all } \Omega$$

and

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)\, e^{j\Omega t}\, d\Omega \quad \text{for all } t$$

In General, $X(j\Omega)$ is a Complex valued function of Ω. Therefore we can write $X(j\Omega)$ as

$$X(j\Omega) = X_R(j\Omega) + jX_I(j\Omega)$$